

Table 1 and the tableone Package

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Key Idea

- Emphasis on summarizing data (both in graphical and tabular formats)
- Frequently instructors do not do a sufficient job explaining relevant summary measures for a particular analysis
- Want some tabular and graphical summaries of potential confounders

Observational Studies

- Observational study: Individuals are not assigned to treatment intervention by an experimental design
- Often unethical or impractical to do randomized trial
- In general, A is not independent of $\{Y^1, Y^0\}$
- Thus, the distribution of $Y|A = 1$ or $Y^1|A = 1$ does not equal the distribution of Y^1 and similarly the distribution of $Y|A = 0$ does not equal the distribution of Y^0
- Why? Confounding
- Heuristically, those who receive treatment may be inherently different than those who do not. Consequently, the associational parameters may reflect such inherent differences as well as any effect of treatment
- In other words, there are common causes of both Y and A

Potential Confounders

- IF the treatment was randomly assigned, then the distribution of any baseline (i.e., pre-randomization) covariate should be the same in each level of the treatment
- Covariates which are related to (i.e., causes of) treatment will be differentially distributed between different levels of the treatment
- Want ways of summarizing the distribution of covariates between treatment levels
- Such a table is often referred to as Table 1

Example of Table 1

(a) Recipient characteristics							
	Deceased donor recipients						
	Maintenance prednisone			Rapid discontinuation of prednisone			P-value
	Median or Frequency	(25th, 75th centile) or percent	Percent Missing	Median or Frequency	(25th, 75th centile) or percent	Percent Missing	
Age (Years)	52.7	(41.2, 61.8)	0	54.8	(43.4, 63.2)	0	<0.001
Race							
Black or African American	26038	32.9	0	7047	28.8	0	<0.001
Hispanic/Latino	12114	15.3		3527	14.4		
Asian	4882	6.2		1568	6.4		
White	34645	43.8		11819	48.3		
Other or Multiracial	1497	1.9		491	2.0		
Gender							
Female	32108	40.6	0	9156	37.4	0	<0.001
Male	47068	59.4		15296	62.6		
Transplant number							

- Many R packages which can create a (pretty) summary table
- TableOne is an R package created by Kazuki Yoshida that eases the construction of “Table 1”, i.e., patient baseline characteristics table commonly found in biomedical research papers.
- This package is also very helpful in providing an overview of your data since it can summarize both continuous and categorical variables mixed within one table.

Some relevant information sources

<https://cran.r-project.org/web/packages/tableone/index.html>

<https://cran.r-project.org/web/packages/tableone/vignettes/introduction.html>

<https://cran.r-project.org/web/packages/tableone/vignettes/smd.html>

http://rstudio-pubs-static.s3.amazonaws.com/13321_da314633db924dc78986a850813a50d5.html

- As an example, we will use the well known iris dataset which contains data on 150 irises including their species type, petal length and width, and sepal length and width

```
CreateTableOne(data=iris)
```

	Overall
n	150
Sepal.Length (mean (SD))	5.84 (0.83)
Sepal.Width (mean (SD))	3.06 (0.44)
Petal.Length (mean (SD))	3.76 (1.77)
Petal.Width (mean (SD))	1.20 (0.76)
Species (%)	
setosa	50 (33.3)
versicolor	50 (33.3)
virginica	50 (33.3)

A New Categorical Variable

Let's add in a new column for color (0 for white, 1 for red) and make the table again:

```
new.iris <- cbind(iris, Petal.Color=rbinom(n=150, size=1, prob=0.5))
CreateTableOne(data=new.iris)
```

	Overall
n	150
Sepal.Length (mean (SD))	5.84 (0.83)
Sepal.Width (mean (SD))	3.06 (0.44)
Petal.Length (mean (SD))	3.76 (1.77)
Petal.Width (mean (SD))	1.20 (0.76)
Species (%)	
setosa	50 (33.3)
versicolor	50 (33.3)
virginica	50 (33.3)
Petal.Color (mean (SD))	0.47 (0.50)

We can see that in this case, the package is unable to determine that Petal.Color is a categorical variable. We will have to set this manually.

Continuous vs Categorical Variables

We can specify which variables we would like in the table, and which variables should be considered categorical:

```
all.vars <- c("Petal.Length", "Petal.Width", "Species", "Petal.Color")
cat.vars <- c("Species", "Petal.Color")
CreateTableOne(vars=all.vars, data=new.iris, factorVars=cat.vars)
```

	Overall
n	150
Petal.Length (mean (SD))	3.76 (1.77)
Petal.Width (mean (SD))	1.20 (0.76)
Species (%)	
setosa	50 (33.3)
versicolor	50 (33.3)
virginica	50 (33.3)
Petal.Color = 1 (%)	70 (46.7)

Notice that Petal.Color only shows one level here.

Print Option 1: Show all levels

The default when a categorical variable has 2 levels is to only show one of the levels. To show all the levels for the categorical variables:

```
t1 <- CreateTableOne(vars=all.vars, data=new.iris, factorVars=cat.vars)
print(t1, showAllLevels = TRUE)
```

	level	Overall
n		150
Petal.Length (mean (SD))		3.76 (1.77)
Petal.Width (mean (SD))		1.20 (0.76)
Species (%)	setosa	50 (33.3)
	versicolor	50 (33.3)
	virginica	50 (33.3)
Petal.Color (%)	0	80 (53.3)
	1	70 (46.7)

Resetting Levels for Categorical Variables

Instead of showing 0/1 for Petal.Color, we can also reset the level for this variable so it displays as “White”/“Red”:

```
new.iris$Petal.Color <- as.factor(new.iris$Petal.Color)
levels(new.iris$Petal.Color) <- c("White", "Red")
t1 <- CreateTableOne(vars=all.vars, data=new.iris)
print(t1, showAllLevels = TRUE)
```

	level	Overall
n		150
Petal.Length (mean (SD))		3.76 (1.77)
Petal.Width (mean (SD))		1.20 (0.76)
Species (%)	setosa	50 (33.3)
	versicolor	50 (33.3)
	virginica	50 (33.3)
Petal.Color (%)	White	80 (53.3)
	Red	70 (46.7)

Notice that if the levels are changed to character values, R will automatically change this variable into a “factor”, you won’t need to manually set it as a categorical variable.

Print Option 2: Interquartile Range

For nonnormal (skewed) continuous variables, the interquartile range (IQR) can be displayed instead of the mean and standard deviation:

```
nn.vars <- c("Sepal.Length", "Sepal.Width")
t2 <- CreateTableOne(data=iris)
print(t2, nonnormal = nn.vars)
```

	Overall
n	150
Sepal.Length (median [IQR])	5.80 [5.10, 6.40]
Sepal.Width (median [IQR])	3.00 [2.80, 3.30]
Petal.Length (mean (SD))	3.76 (1.77)
Petal.Width (mean (SD))	1.20 (0.76)
Species (%)	
setosa	50 (33.3)
versicolor	50 (33.3)
virginica	50 (33.3)

Checking Missingness

The TableOne package can also help you check for missingness in your data. Here we set the first 10 Petal.Length values to NA and check for missingness:

```
na.iris <- iris; na.iris$Petal.Length[1:10] <- rep(NA, 10)
t3 <- CreateTableOne(data=na.iris)
summary(t3)
```

```
### Summary of continuous variables ###
```

```
strata: Overall
```

	n	miss	p.miss	mean	sd	median	p25	p75	min	max	skew	kurt
Sepal.Length	150	0	0	6	0.8	6	5.1	6	4.3	8	0.3	-0.6
Sepal.Width	150	0	0	3	0.4	3	2.8	3	2.0	4	0.3	0.2
Petal.Length	150	10	7	4	1.7	4	1.7	5	1.0	7	-0.4	-1.2
Petal.Width	150	0	0	1	0.8	1	0.3	2	0.1	2	-0.1	-1.3

```
#####  
### Summary of categorical variables ###
```

```
strata: Overall
```

var	n	miss	p.miss	level	freq	percent	cum.percent
Species	150	0	0.0	setosa	50	33.3	33.3
				versicolor	50	33.3	66.7
				virginica	50	33.3	100.0

Stratify by Group and Testing

Often times you want to see summaries of different strata, with the strata usually being the treatment variable. Going back to our Petal.Color example:

```
CreateTableOne(data = new.iris, strata = "Petal.Color")
```

	Stratified by Petal.Color			
	White	Red	p	test
n	80	70		
Sepal.Length (mean (SD))	5.92 (0.84)	5.76 (0.82)	0.234	
Sepal.Width (mean (SD))	3.07 (0.43)	3.04 (0.44)	0.733	
Petal.Length (mean (SD))	3.87 (1.78)	3.63 (1.76)	0.403	
Petal.Width (mean (SD))	1.23 (0.76)	1.16 (0.77)	0.585	
Species (%)			0.512	
setosa	25 (31.2)	25 (35.7)		
versicolor	30 (37.5)	20 (28.6)		
virginica	25 (31.2)	25 (35.7)		
Petal.Color = Red (%)	0 (0.0)	70 (100.0)	<0.001	

Note the group comparison p-value that are printed in the table. The hypothesis test functions used by default are `chisq.test()` for categorical variables (with continuity correction) and `oneway.test()` for continuous variables (with equal variance assumption, i.e., regular ANOVA).

Two-group ANOVA is equivalent of t-test.

Print Option 3: Standardized Mean Differences

The Standardized Mean Difference (SMD), also known as Cohen's D, is a measure of distance between two group means in terms of one or more variables. In practice it is often used as a balance measure of individual covariates before and after propensity score matching. (More on this later in the semester...) Using our example from the previous slide:

```
t3 <- CreateTableOne(data = new.iris, strata = "Petal.Color")
print(t3, smd = TRUE)
```

	Stratified by Petal.Color		p	test SMD
	White	Red		
n	80	70		
Sepal.Length (mean (SD))	5.92 (0.84)	5.76 (0.82)	0.234	0.196
Sepal.Width (mean (SD))	3.07 (0.43)	3.04 (0.44)	0.733	0.056
Petal.Length (mean (SD))	3.87 (1.78)	3.63 (1.76)	0.403	0.137
Petal.Width (mean (SD))	1.23 (0.76)	1.16 (0.77)	0.585	0.089
Species (%)			0.512	0.191
setosa	25 (31.2)	25 (35.7)		
versicolor	30 (37.5)	20 (28.6)		
virginica	25 (31.2)	25 (35.7)		
Petal.Color = Red (%)	0 (0.0)	70 (100.0)	<0.001	NaN

Print Option 4: Variable Name Labels

Using the “labelled” package, a label can be assigned to each variable and displayed in the final printed table. This way, your variable names stay intact for further manipulations in your code but your tableone output will be presentation quality:

```
new.iris <- set_variable_labels(new.iris,  
                                Sepal.Length = "Length of Sepal",  
                                Sepal.Width = "Width of Sepal",  
                                Petal.Length = "Length of Petal",  
                                Petal.Width = "Width of Petal",  
                                Petal.Color = "Color of Petal"  
)  
print(CreateTableOne(data=new.iris), varLabels = TRUE)
```

	Overall
n	150
Length of Sepal (mean (SD))	5.84 (0.83)
Width of Sepal (mean (SD))	3.06 (0.44)
Length of Petal (mean (SD))	3.76 (1.77)
Width of Petal (mean (SD))	1.20 (0.76)
Species (%)	
setosa	50 (33.3)
versicolor	50 (33.3)
virginica	50 (33.3)
Color of Petal = Red (%)	70 (46.7)

In my opinion, a huge advantage of using the `tableone` package is how easily this R table can be exported. To turn this table into LATEX code, we use the “`xtable`” package here:

```
final.table <- print(CreateTableOne(data=new.iris), varLabels = TRUE)
print(xtable(final.table))
```

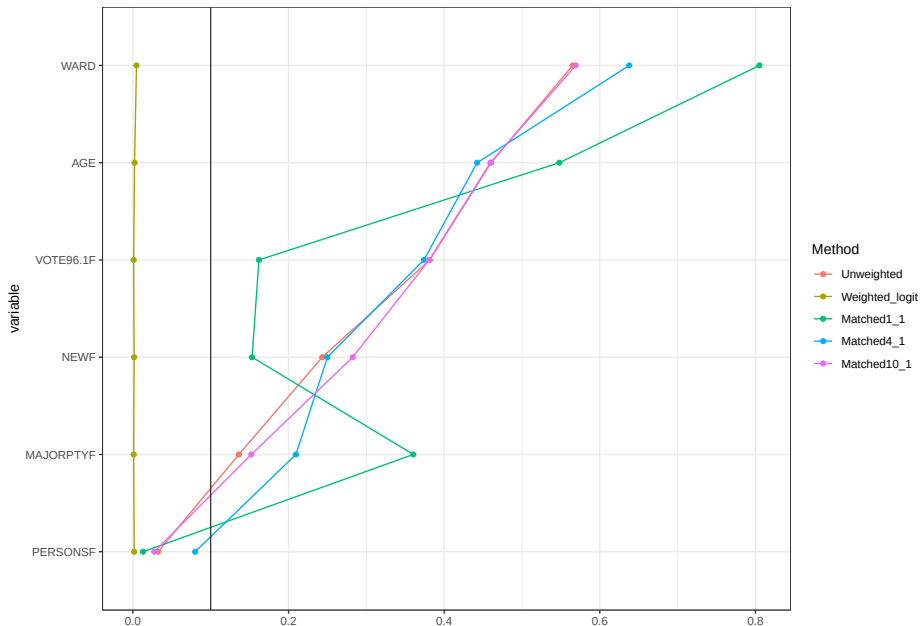
You can just as easily export this table to a CSV format (for Excel):

```
write.csv(final.table, file = "myTable.csv")
```

Plot of SMD

- If we have a lot of baseline covariates it is helpful to plot the absolute value of the standardized mean difference
- We will discuss methods in the future (e.g., weighting, matching, etc.) which help reduce the imbalance (i.e., absolute SMD) among groups. Plotting those on the same plot can show how we “fixed” the imbalance.
- Example given on the following slide of the SMD in the “original” dataset and then under various matching and weighting methods

Plot of SMD



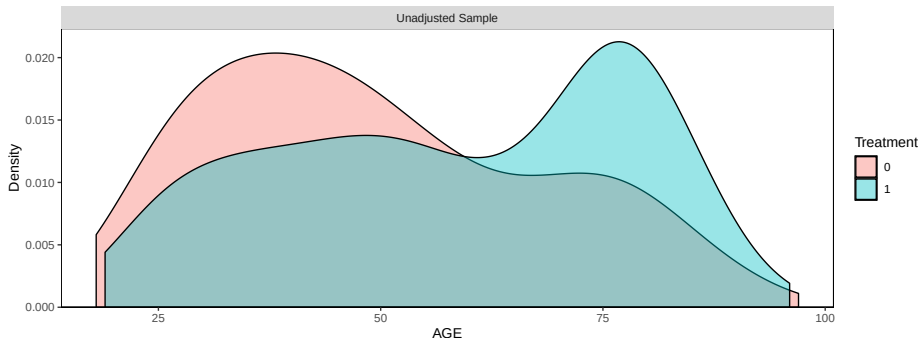
Other plots

- Can be useful to assess overlap and similarity in the distribution of covariates; will introduce more formally in the context of matching, weighting
- Use cobalt package in R

Overlap and Similarity of Age

```
library(cobalt)
bal.plot(as.formula(PHN.C1 ~ (PERSONS + VOTE96.1 + NEW + MAJORPTY +
  AGE + WARD)), data = imai, "AGE", which = "unadjusted")
```

Distributional Balance for "AGE"



Practice

Load the following dataset from the Mayo Clinic trial in primary biliary cirrhosis of the liver:

```
library(survival)
data(pbc)
```

Try answering the following questions using the pbc dataset:

- How many male patients are there in the dataset, and what is the percentage?
- Turn `c("status", "trt", "ascites", "hepato", "spiders", "edema", "stage")` into categorical variables and show all levels. What percent of patients are stage 3?
- How many missing values are there for "copper" and "stage"? What percent of values are missing for "trig"?
- Display the IQR for `c("bili", "chol", "copper", "alk.phos", "ast", "trig", "protime")`. What is the median for "chol"? What is the 25th percentile value for "ast"?